Highlights

Continuum Memory Architecture (CMA) for Clinical Search: A Simulation-Based Evaluation of Intent-Aware Retrieval Using Complex Public Clinical Data

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- CMA's JEPA-driven prefetching and GSI gating reduce clinical search latency by 21–23%.
- CMA reduces time-to-correct-information by 13–17% with 100% task completion.
- SPD manifold intent encoding tracks topic pivots via geodesic shifts.
- The advantage is stable across 1–10,000 vignettes (3–30,000 sessions).
- CMA lowers simulated cognitive load by 16–18% versus sparse baselines.